Report 1: Project Description Report

Hotel Booking System

Course / Assignment	UML Class Diagram Modelling Assignment
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1. Project Title and Business Idea

Project title: Hotel Booking System

Business idea: A mid-sized hotel needs an information system to manage its guests, rooms, room categories, reservations and payments. Guests can reserve rooms for chosen dates, staff members process and confirm reservations, and every booking is linked to the payments made for it.

2. Problem Statement and Purpose

Many hotels still record reservations in paper registers or scattered spreadsheets. This causes **double bookings**, lost guest records, slow check-in, unclear payment histories and difficulty knowing which rooms are free on a given date. Staff waste time searching for information, and guests receive inconsistent service.

Purpose: to design a clear, structured model (UML class diagram) of a Hotel Booking System that centralises guest, room, booking and payment information, so that a future application can prevent double bookings, track payments and speed up hotel operations.

3. Project Scope

3.1 Functions covered (in scope)

• Registering and storing guest information.

• Managing rooms and room types (price, capacity, status).

• Checking room availability for a date range.

• Creating, modifying, confirming and cancelling bookings.

• Recording payments, issuing receipts and processing refunds.

• Staff handling of bookings, room assignment, check-in and check-out.

3.2 Functions excluded (out of scope)

• Developing a fully working software application (this project is a design/modelling task only).

• Online payment gateway or mobile-money integration.

• Restaurant, spa, housekeeping, inventory and staff payroll management.

• Accounting, tax reporting and multi-branch hotel chains.

• User interface design and database implementation.

4. Main Classes and Their Responsibilities

Class	Responsibility
Guest	Stores guest details; creates, views and cancels bookings.
Booking	Central record of a reservation: dates, status and total cost.
Room	Represents a physical hotel room and its availability.
Staff	Hotel employee who processes bookings, check-in and check-out.
Payment	Records money paid for a booking and issues receipts/refunds.
RoomType	Category of room (Single, Double, Suite) with price and capacity.

5. System Requirements

5.1 Functional requirements

• The system shall store guest details (name, email, phone, national ID).

• The system shall store each room with its number, floor, status and room type.

• The system shall let a guest make a booking for a room with check-in and check-out dates.

• The system shall prevent two confirmed bookings for the same room on overlapping dates.

• The system shall calculate the total booking amount from the room type's price per night and the length of stay.

• The system shall record one or more payments per booking and generate receipts.

• The system shall let staff confirm bookings, assign rooms, and check guests in and out.

• The system shall allow bookings to be modified or cancelled, with refunds where applicable.

5.2 Non-functional requirements

• **Usability:** simple enough for hotel staff with basic computer skills.

• **Reliability:** booking and payment data must stay consistent and not be lost.

• **Security:** only authorised staff may confirm bookings or process refunds; guest data must be protected.

• **Performance:** availability searches should return results within a few seconds.